Applications of quadratic functions

1

a $t = 2, t = 3$

b 5 sec

c No

2

a $t = \frac{1}{4}, t = 2\frac{3}{4}$

b $2\frac{1}{2} \text{ sec}$

c $t = 1\frac{1}{4}, t = 1\frac{3}{4}$

d $\frac{1}{2} \text{ sec}$

3

a $w = 10$

b 16 yards

4 $t = 4$

5

a $x = 10$

b Square of the number is 100.
17 times the number is 170.
$170 - 100 = 70$

6

a 6 in

b 6.6 in

7

a $4x^2 + 44x \text{ m}^2$

b $x = 2$

8

a $m = 10$

b 15 hours

9

a $w = 2.33$

b 15.3 ft

10

a 120

b The amount of time it takes for the frisbee to reach the ground.

c 160 meters

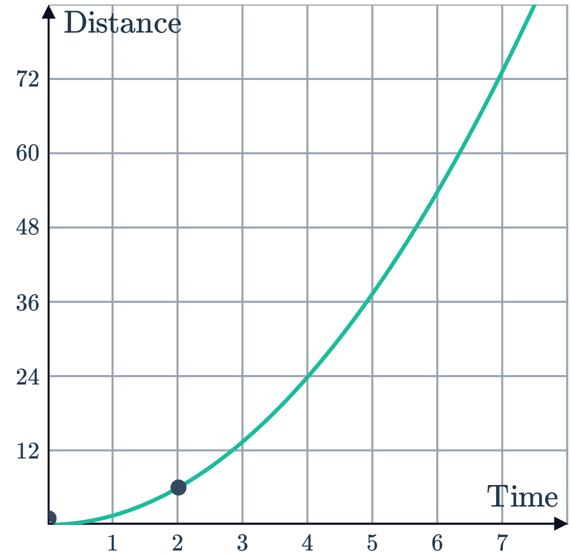
11 3

12

a

Time	0	2	4	6
Distance	0	6	24	54

b



c $t = 2$

13

a 313.6 yards

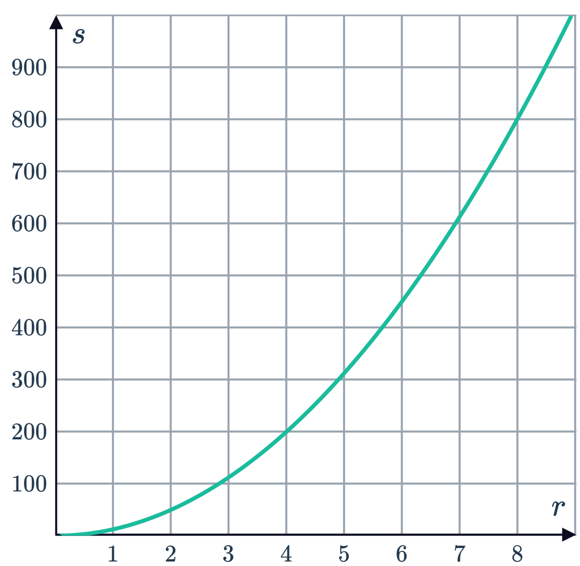
b $t = 4$

14

a

r	1	2	3	4	5	6	7	8
S	12.57	50.27	113.1	201.06	314.16	452.39	615.75	804.25

b



c

i $S = 380$

ii $r = 4$

15

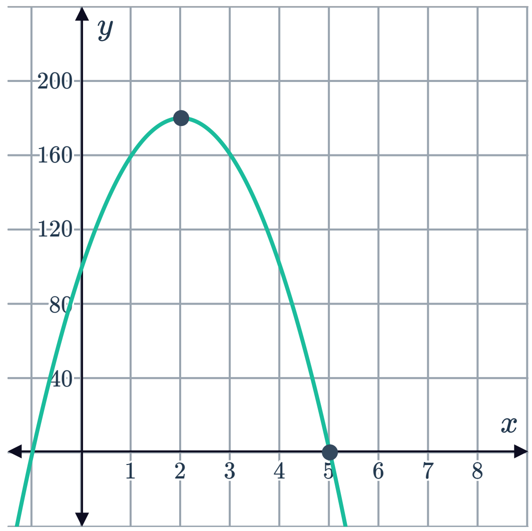
a

x	0	1	2	3	4	5	6
y	100	160	180	160	100	0	-140

b 5 seconds

c 180 feet

d



16

a $y=-\frac{1}{20}(x-18)^2+16.2$

b $4071.50\ \text{yd}^2$

17

a $A(0,3.5), B(3,0), C(7,0)$

b 3

c $9a+k=\frac{7}{2}$

d $0=16a+k$

e $a=-\frac{1}{2}$

f $8\ \text{m}$

18

a $A(18,0), B(0,108)$

b $y=-\frac{1}{3}(x-18)(x+18)$

c $x=9\sqrt{2}$

19

a $(180,98)$

b $y=-\frac{49}{16\,200}(x-180)^2+98$

c Below

20

a

t	0	1	2	3	5	6
d	0	16	64	144	400	576

b Initially, the object has not traveled any distance.

c Because time cannot be negative.

21

a $t = 3$

b $t = 9$

22 $t = 6$

23

a $m = 1900$

b 64.8 km/h

c $v = 16$

24

a $S = 528$

b $n = 5$

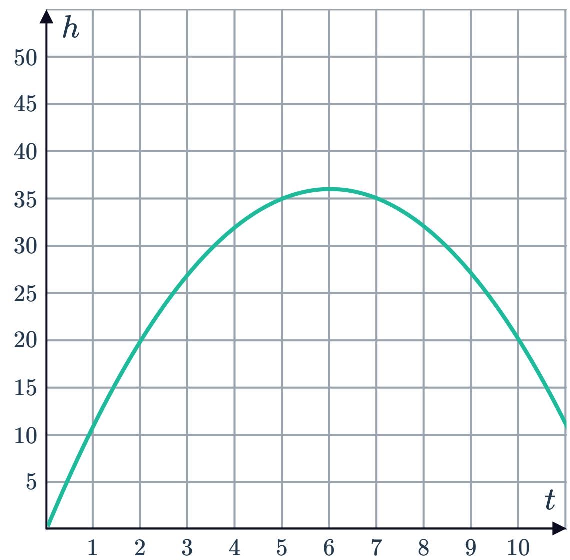
Maximum and minimum problems

25

a

t	1	2	3	4	5	6	7	8	9	10
h	11	20	27	32	35	36	35	32	27	20

b



c 33.75 meters

d 36 meters

26

a 58 **b** $80 - x$ **c** $y = x(80 - x)$ **d** 1600

e Yes

27

a 336 square meters **b** 32 m

c 400 square meters **d** Length $= 20 \text{ m}$, Width $= 20 \text{ m}$

28

a $y = x(24 - x)$ **b** $x = 12$ **c** 144

29

a $A = x(20 - x)$ **b** 10 units **c** 10 units

30

a $s = 12$

b 337

31

a $t = 3$

b 122 units

32

a $t = 2.84 \text{ s}$

b $t = 1.38 \text{ s}$

c 34.25 m

33

a $13 - x$

b $x(13 - x)$

c 6.5 yd

d 42.25 yd^2

34

a $(168 - x) \text{ m}$

b $x(168 - x) \text{ m}^2$

c 84 m

d 7056 m^2

35

a 16

b 64 meters

c 16 meters

d 39 meters

e $0 \leq d \leq 32$

36

a 576 feet

b 6 seconds

c 12 seconds

37

a 50 meters

b $x = -10, 10$

c 37.50 meters

d $x < -20$ and $x > 20$